
Archelec Manifesto Analysis: Discriminative Classification and Semantic Topology

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Abstract

This study presents an empirical comparative analysis of Natural Language Processing (NLP) architectures applied to historical French political discourse. Utilizing the Archelec corpus of legislative manifestos (1981, 1988 and 1993 elections, $N = 8, 110$), we contrast sparse discriminative baselines against dense contextual language models for the task of political affiliation classification. We demonstrate that a TF-IDF feature space optimized via a Linear Support Vector Machine (SVM) achieves superior discriminative performance (Macro- $F_1 = 0.831$), outperforming a supervised CamemBERT Linear Probe and rendering Zero-Shot NLI inference inadequate for this highly localized domain. Beyond discrete classification, we map the global semantic topology using zero-mean isotropic CamemBERT centroids ($\mathbb{R}^{768}$), revealing structural proximities—such as the severe convergence of the RIGHT_UDF and RIGHT_RPR ($\cos = 0.822$)—that mathematically explain the linear boundary degradations observed in the baseline. Finally, we implement a strict topological hygiene pipeline, combining DBSCAN ϵ -deduplication and Local Outlier Factor (LOF) scoring to isolate extreme intra-class anomalies, proving the necessity of ambient space evaluation over 2D non-linear projections. This framework provides a rigorous methodological benchmark for mapping ideological boundaries and semantic overlaps in fragmented political text.

1 Introduction

The quantitative analysis of political discourse is a foundational task in computational social science. The digitized Archelec corpus Pélissier et al. [2018], containing exhaustive electoral manifestos from the French Fifth Republic, provides a robust medium to explore the semantic structures of political rhetoric. However, applying Natural Language Processing (NLP) to historical political corpora introduces distinct methodological challenges: extreme multi-class imbalance, localized administrative vocabulary, and the inherent tension between pragmatic electoral alliances and underlying ideological proximities.

This project serves as an empirical exploration of these challenges, comparing the efficacy of distinct NLP architectures in modeling French political text. Specifically, we evaluate whether large pre-trained contextual embeddings (Transformer architectures) inherently supersede traditional sparse lexical baselines (TF-IDF) in a highly localized classification task.

Our approach is bifurcated into two primary objectives: discriminative boundary extraction and generative semantic mapping. First, we model political affiliation as a discrete multi-class problem ($f : \mathcal{X} \rightarrow \mathcal{Y}$), benchmarking a sparse Linear Support Vector Machine (SVM) against a CamemBERT Linear Probe and a Zero-Shot mDeBERTa Natural Language Inference (NLI) model. We aim to determine which representation space, sparse discrete tokens ($\mathbb{R}^{18,054}$) or dense contextual embeddings ($\mathbb{R}^{768}$), yields the optimal decision boundary.

Second, acknowledging that rigid classification boundaries frequently obscure shared vocabularies, we transition from predictive modeling to unsupervised topological mapping. By analyzing the isotropic centroids of the contextual embeddings and deploying density-based outlier detection algorithms (LOF), we cross-validate the failure states of the discriminative models.

The core contributions of this exploratory study are:

1. **Architectural Benchmarking:** An empirical demonstration that sparse TF-IDF representations out-perform contextualized LLMs for resolving localized political affiliations where decision boundaries rely heavily on named entities and explicit alliance markers.
2. **Semantic Proximity Analysis:** The use of zero-mean class centroids to quantify hidden ideological overlaps (e.g., the convergence of the fragmented center-right blocs), effectively explaining the off-diagonal leakage observed in the SVM confusion matrix.
3. **Topological Hygiene:** A rigorous demonstration of the limitations of 2D non-linear dimensionality reduction (UMAP) for outlier detection, proving the necessity of ambient-space density evaluation to identify genuine semantic anomalies.

2 Corpus architecture and pre-processing

2.1 The Archelec corpus and diachronic label alignment

We utilize the Archelec corpus, a digitized archive of French electoral manifestos compiled by the CEVIPOF (Sciences Po) Pélissier et al. [2018]. To ensure structural homogeneity and maintain sufficient statistical power, we restrict our analysis to the legislative elections of 1981, 1988, and 1993, yielding $N = 8,110$ documents. Presidential manifestos were excluded due to their divergent rhetorical formatting and low sample size."

The official metadata records the political affiliation (*titulaire-soutien*) of each candidate. However, raw administrative labels suffer from sparsity and temporal fragmentation. To establish a stable classification objective $f : \mathcal{X} \rightarrow \mathcal{Y}$, we aggregate these raw labels into ideological macro-blocs. The Left and Far-Right are mapped globally across all years: LEFT_PCF (Communist), LEFT_PS (Socialist/Radical), FAR_LEFT, ECO (Greens), and FAR_RIGHT_FN (National Front).

Conversely, the mainstream Right undergoes diachronic alignment to reflect conditional electoral alliances between the RPR and UDF. When explicit joint lists occur, they are mapped to their historical coalition: RIGHT_UNM (1981), RIGHT_URC (1988), and RIGHT_UPF (1993). Independent candidacies remain isolated as RIGHT_RPR or RIGHT_UDF. We explicitly note a critical limitation regarding our target variable definition: the ideological macro-blocs are subjective, rather than official governmental classifications (such as the *nuances politiques* issued by the Ministère de l'Intérieur). Consequently, these labels do not guarantee genuine ideological proximity or strict semantic homogeneity within the aggregated macro-blocs. High intra-class variance is structurally expected, as candidates frequently adopt localized rhetoric that deviates significantly from cohesive national party platforms.

As illustrated in Figure 1, the global corpus geometry exhibits severe multi-class imbalance. The continuous left-wing and ecological factions dominate the dataset (LEFT_PCF: $N = 1,605$; LEFT_PS: $N = 1,563$), followed by the FAR_RIGHT_FN ($N = 1,242$).

This skew is primarily a structural artifact of our diachronic label alignment rather than pure electoral participation. Factions like the LEFT_PCF maintain continuous, independent electoral presence across all three cycles (1981, 1988, 1993), accumulating larger aggregate sample sizes. Conversely, the mainstream Right's reliance on conditional, year-specific coalitions fragments their representation. For example, RIGHT_UNM ($N = 221$) and RIGHT_URC ($N = 493$) are bounded to 1981 and 1988, respectively, artificially capping their global frequencies.

The temporal distribution (Figure 1, right) further exposes historical volatility. The anomalous density of the LEFT_PS in 1981 directly reflects the legislative mobilization following François Mitterrand's presidential victory, a momentum that normalizes in subsequent electoral cycles. This extreme empirical imbalance—where the majority class outnumbers the minority by a factor of 7.2—strictly necessitates the inverse-frequency loss scaling ($\mathcal{L}_{WCE}$) applied during the optimization of our discriminative architectures (Section 3).

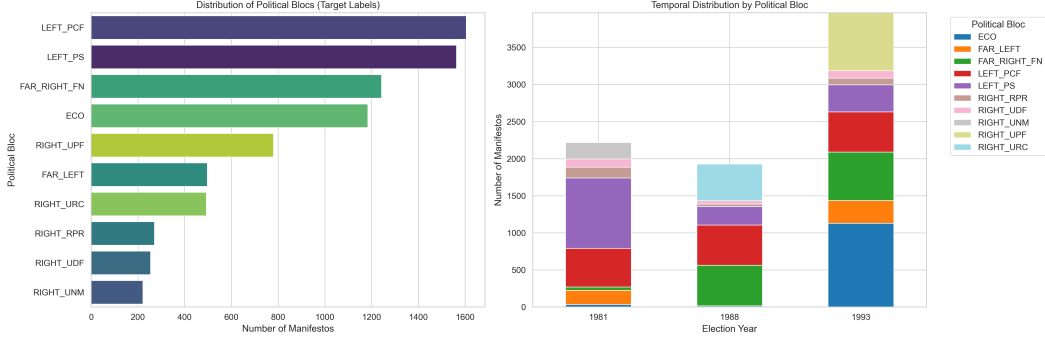


Figure 1: Global and temporal distribution of political manifestos across the Archelec corpus ($N = 8,110$). The structural fragmentation of the right-wing macro-blocs is an artifact of conditional, year-specific electoral alliances.

2.2 Lexical normalization and dimensionality reduction

The raw OCR transcriptions contain noise, administrative boilerplate, and geographical indicators. If unhandled, a linear classifier will exploit spatial auto-correlation (e.g., predicting political blocs based on regional commune names rather than ideological discourse). We implement a strict Natural Language Processing (NLP) pipeline via `spaCy` to isolate the ideological signal:

1. **Heuristic sanitization:** Regular expressions remove archival boilerplate (e.g., “Fonds CEVIPOF”, printer metadata) and standardize common acronyms (e.g., “P.S.”, “R.P.R.”).
2. **Morphological and syntactic filtering:** We restrict the vocabulary to semantically dense Parts-of-Speech (POS): NOUN, ADJ, VERB, and PROP. Tokens are lemmatized, and diacritics are stripped via Unicode NFKD normalization to mitigate OCR-induced accent-dropping. Simple negations are preserved by appending a `ne_` prefix to the subsequent lemma.
3. **Spatial and discourse masking:** We cross-reference tokens against a deterministic geographical exclusion set derived from the 2025 INSEE Code Officiel Géographique (COG) gazetteer. Furthermore, domain-specific discourse markers devoid of ideological variance (e.g., *circonscription*, *bulletin*, *candidat*) are purged (see Appendix 5 for the exhaustive exclusion criteria).

This pipeline induces an aggressive but methodologically necessary compression. The average manifesto length is reduced from 651 raw words to 194 clean tokens, effectively removing 70.5% of the raw text as noise. The resulting normalized global vocabulary space is reduced to $|\mathcal{V}| = 31,747$ unique dimensions.

2.3 Corpus partitioning and class stratification

To rigorously evaluate the generalization bounds of our discriminative architectures, we partition the fully pre-processed corpus $\mathcal{D}$ into training and testing sets ($\mathcal{D}_{train} \cap \mathcal{D}_{test} = \emptyset$). We apply an 80/20 randomized hold-out split, yielding $|\mathcal{D}_{train}| = 6,488$ and $|\mathcal{D}_{test}| = 1,622$ manifestos. Given the multi-class imbalance, we enforce target stratification. This ensures that the empirical class distribution $P(\mathcal{Y})$ is preserved across both subsets, a necessary condition for unbiased model evaluation.

3 Manifestos classification

Having standardized the corpus into a normalized text space, we formalize the extraction of political affiliation as a supervised classification problem. Our objective is to learn a mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X}$ is the document representation and $\mathcal{Y} \in \{C_1, \dots, C_k\}$ represents the discrete ideological macro-blocs.

To rigorously evaluate the optimal representation space for political discourse, we compare a sparse discriminative baseline (TF-IDF + Linear SVM Joachims [1998]) against two advanced contextualized architectures: a Zero-Shot inference model (mDeBERTa He et al. [2021]) and a supervised Linear Probe trained on frozen Transformer embeddings (CamemBERT Martin et al. [2020]).

3.1 Sparse lexical baseline: TF-IDF and Linear SVM

Traditional political text analysis relies heavily on word frequencies. To establish a robust baseline, we map the pre-processed manifestos to a sparse feature space $\mathbb{R}^{|\mathcal{V}|}$ using Term Frequency-Inverse Document Frequency (TF-IDF). After filtering for document frequency boundaries (removing corpus-specific stop-words and highly idiosyncratic typos), we extract a vocabulary of $|\mathcal{V}| = 18,054$ uni-grams and bi-grams.

This extraction yields a sparse representation, with the training matrix exhibiting a sparsity of 98.66%. To prevent magnitude dominance from varying manifesto lengths, vectors are L_2 -normalized (the exact mathematical formulation and parameterization of the TF-IDF space are detailed in Appendix B).

Given the high dimensionality and extreme sparsity of $\mathbb{R}^{18,054}$, non-linear algorithms are highly susceptible to overfitting. Therefore, we optimize a One-vs-Rest (OvR) Linear Support Vector Machine (SVM). The linear SVM is mathematically ideal for this space, as it seeks the optimal hyperplane separating class i from $\neg i$ by maximizing the margin:

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i \xi_i \quad (1)$$

To prevent the objective function from trivially optimizing toward the majority political blocs (e.g., mainstream coalitions) at the expense of minority factions, we modify the standard hinge loss formulation to incorporate inverse-frequency class weights. The penalty parameter C is adjusted per class j , such that $C_j = C \times \frac{N}{K \cdot n_j}$, where n_j is the empirical frequency of class j . The balanced objective becomes:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{j=1}^K C_j \sum_{i \in \mathcal{C}_j} \xi_i \quad (2)$$

The optimization of the regularization hyperparameter C was conducted via 5-Fold Stratified Cross-Validation over the training partition ($\mathcal{D}_{train}$). To ensure the preservation of minority class performance during grid search, the optimization target was explicitly set to the macro-averaged F_1 score rather than raw accuracy. The search space explored logarithmic bounds of $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 10^0, 5, 10^1\}$.

Beyond establishing a robust baseline performance metric, the optimized SVM’s weight matrix $\mathbf{W} \in \mathbb{R}^{\tilde{C} \times V}$ provides an explicitly interpretable ledger. By extracting the highest positive coefficients $\mathbf{w}_i$ mapping to the sparse TF-IDF vocabulary, we empirically isolate the most polarizing discriminative tokens defining each political faction’s rhetoric.

3.2 Contextual zero-shot inference via NLI (mDeBERTa)

While the sparse SVM baseline effectively isolates high-frequency discriminative tokens, it is inherently blind to polysemy, syntax, and implicit semantic contexts. To evaluate the zero-shot generalization capabilities of pre-trained contextualized language models on French political discourse, we implement a Natural Language Inference (NLI) architecture.

Rather than updating model weights to learn a direct mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$, NLI recasts multi-class classification as a sequence-pair entailment task. We utilize mDeBERTa-v3, a multilingual transformer fine-tuned on the MNLI and XNLI datasets. For a given manifesto sequence x (the premise) and a candidate ideological label $y \in \mathcal{Y}$, we construct a natural language hypothesis h_y using a deterministic template: “*Ce manifeste politique soutient [label_description].*”

Because the raw archival labels (e.g., FAR_RIGHT_FN) lack inherent semantic meaning to the pre-trained model, we map each target class to a descriptive linguistic anchor (e.g., “*le Front National et*

l'extrême droite”, see Appendix C for the exhaustive mapping framework). The model computes the entailment logit $s_{\text{entail}}(x, h_y)$, and the final classification probability over the discrete political factions is obtained via a softmax normalization over all candidate hypotheses:

$$P(y \mid x) = \frac{\exp(s_{\text{entail}}(x, h_y))}{\sum_{y' \in \mathcal{Y}} \exp(s_{\text{entail}}(x, h_{y'}))} \quad (3)$$

Due to the quadratic complexity of the self-attention mechanism, the transformer’s context window is strictly bounded to 512 tokens. While our pre-processing compression (Section 2) reduces the average manifesto to 194 tokens, we acknowledge that ideological signals located at the trailing edge of exceptionally long documents are structurally truncated by this architecture.

3.3 Contextual supervised classification: CamemBERT linear probe

The Zero-Shot NLI framework evaluates the geometric alignment of pre-trained embeddings but cannot optimize for corpus-specific ideological boundaries. To maximize discriminative performance while retaining contextual awareness, we optimize a Linear Probe on top of a French-native Transformer (almanach/camembert-base).

We freeze all internal attention layers (the roberta encoder), restricting optimization to a newly initialized linear classification head: $f : \mathbb{R}^{768} \rightarrow \mathbb{R}^C$. To optimize this head, we partition $\mathcal{D}_{\text{train}}$ further, extracting an internal 10% validation set ($\mathcal{D}_{\text{val}}$) to govern Early Stopping (patience=5 epochs) without leaking the final test set $\mathcal{D}_{\text{test}}$. To mitigate the multi-class imbalance, the standard Cross-Entropy objective is scaled by inverse class frequencies, yielding the Weighted Cross-Entropy Loss ($\mathcal{L}_{WCE}$):

$$\mathcal{L}_{WCE} = - \sum_{i=1}^N \sum_{c=1}^C w_c y_{i,c} \log(\hat{y}_{i,c}) \quad (4)$$

where $w_c = \frac{N}{C \cdot n_c}$.

The model is trained for a maximum of 15 epochs using the AdamW optimizer with a learning rate of 10^{-3} and a weight decay of 0.01. The best set of weights, evaluated against the macro- F_1 score on $\mathcal{D}_{\text{val}}$, is restored before the final out-of-sample inference on $\mathcal{D}_{\text{test}}$.

3.4 Empirical Results and models evaluation

We evaluate the discriminative capacity of our three architectures on the hold-out test set ($\mathcal{D}_{\text{test}}$, $N = 1,622$). To account for the severe class imbalance inherent to the dataset, our primary optimization and evaluation metric is the unweighted macro-averaged F_1 score:

$$\text{Macro-}F_1 = \frac{1}{C} \sum_{i=1}^C 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (5)$$

As detailed in Table 1, the sparse linear baseline (TF-IDF + SVM) significantly outperforms both contextual transformer architectures. The SVM achieves a Macro- F_1 of 0.831, exceeding the CamemBERT Linear Probe by a margin of $\Delta = +0.109$.

Table 1: Global Classification Performance ($\mathcal{D}_{\text{test}}$). Bold values indicate the optimal score across architectures. All metrics are macro-averaged to adjust for class imbalance.

Architecture	Precision	Recall	Macro- F_1	Accuracy
TF-IDF + Linear SVM	0.852	0.823	0.831	0.925
CamemBERT (Linear Probe)	0.738	0.733	0.722	0.858
mDeBERTa (Zero-Shot NLI)	0.408	0.378	0.312	0.430

The catastrophic failure of the Zero-Shot NLI model (Macro- $F_1 = 0.312$) demonstrates that pre-trained multilingual representations natively lack the localized, diachronic nuance necessary to resolve fine-grained French political alliances (e.g., distinguishing the RPR from the UDF). While

training a supervised Linear Probe over the CamemBERT embeddings significantly bridges this deficit by mapping the contextual manifold to the specific electoral targets, it still fails to surpass the baseline. Ultimately, this establishes a fundamental principle of task-model coherence: the simplest architecture, requiring orders of magnitude less compute while providing explicit lexical interpretability, yields the optimal discriminative boundary.

Consequently, we establish that the sparse $\mathbb{R}^{|V|}$ representation is strictly superior for discriminative boundary extraction. We relegate the exhaustive, per-class classification matrices for all models to Appendix D.

3.5 Discriminative feature analysis

While black-box architectures obscure the exact locus of decision-making, the Linear SVM provides an explicitly interpretable weight matrix $\mathbf{W} \in \mathbb{R}^{10 \times 18,054}$. In a One-vs-Rest (OvR) topology, the highest positive coefficients $\mathbf{w}_i$ isolate the tokens that most aggressively maximize the margin for class i , whereas the lowest negative coefficients identify tokens whose presence strongly predicts that the document belongs to a competing manifold.

Table 2 extracts the top 3 positive and negative features defining the optimal hyperplanes for each ideological macro-bloc.

Table 2: Top discriminative SVM bounds. Positive weights define the class manifold; negative weights enforce the boundary against competing manifolds.

Political Bloc	Top Positive Features (+w)	Top Negative Features (-w)
FAR_LEFT	lcr (+1.24), enfant fonctionnaire (+1.03), gauche vraiment (+0.92)	medaille commemoratif (-0.54), commemoratif (-0.52), legislative eme (-0.52)
LEFT_PCF	communiste (+4.24), parti communiste (+3.37), droite (+1.91)	socialiste (-1.54), parti socialiste (-1.16), rpr (-1.13)
LEFT_PS	socialiste (+2.66), parti socialiste (+2.39), radical socialiste (+2.34)	communiste (-4.50), union (-3.38), parti communiste (-3.13)
ECO	ecologie (+3.44), ecologiste (+2.58), solidarite ecologie (+1.27)	autogestion (-0.89), rpr (-0.82), nante (-0.81)
RIGHT_UNM	rpr soutenir (+2.59), union majorite (+2.29), majorite presenter (+2.00)	rassemblement (-1.29), apporter (-1.27), union rassemblement (-1.04)
RIGHT_URC	union rassemblement (+4.08), rassemblement centre (+2.72), urc (+2.11)	changement (-1.91), union majorite (-1.45), dominiqu (-1.42)
RIGHT_UDF	udf (+2.75), union democratie (+2.74), sortir udf (+1.79)	rpr (-3.01), rpr udf (-2.32), rassemblement republique (-2.05)
RIGHT_RPR	rassemblement republique (+4.42), union majorite (+2.14), rpr (+2.13)	udf (-3.10), union rpr (-2.51), rpr udf (-1.74)
RIGHT_UPF	union rpr (+4.34), union udf (+2.54), union (+2.51)	union rassemblement (-2.65), liberte (-1.93), rassemblement centre (-1.79)
FAR_RIGHT_FN	immigre (+1.06), immigration (+1.03), patrie (+1.02)	licencier droit (-1.13), administrateur (-0.82), enfant fonctionnaire (-0.80)

The structural utility of the negative bounds is immediately apparent when analyzing adjacent factions. The hyperplane defining the LEFT_PS heavily penalizes the tokens *communiste* (-4.50) and *parti communiste* (-3.13). This does not indicate that the Socialists are ideologically attacking the Communists; rather, it mathematically demonstrates that the presence of these tokens is the primary mechanism by which the SVM prevents a Communist manifesto from breaching the Socialist

decision boundary. Similarly, the RIGHT_RPR boundary is enforced by strongly penalizing udf (-3.10), isolating the independent Gaullists from their centrist rivals.

3.6 Multi-class boundary degradation and off-diagonal density

To evaluate the limitations of our optimal discriminative baseline, we analyze the multi-class Area Under the Receiver Operating Characteristic Curve (AUC-ROC) alongside the empirical confusion matrix M on the hold-out test set ($\mathcal{D}_{test}$). Let $M_{i,j}$ denote the frequency of manifestos belonging to true ideological bloc i that are classified as bloc j .

As illustrated in Figure 2, the linear boundaries perfectly isolate the ideological extremes. The FAR_RIGHT_FN and FAR_LEFT achieve near-perfect discrimination ($AUC = 1.00$). Interestingly, the LEFT_PCF exhibits zero off-diagonal overlap with the FAR_LEFT ($M_{PCF,FAR_LEFT} = 0$), instead leaking slightly into the mainstream LEFT_PS ($M_{PCF,PS} = 8$), reflecting localized pragmatic alliances between Communists and the incumbent Socialists.

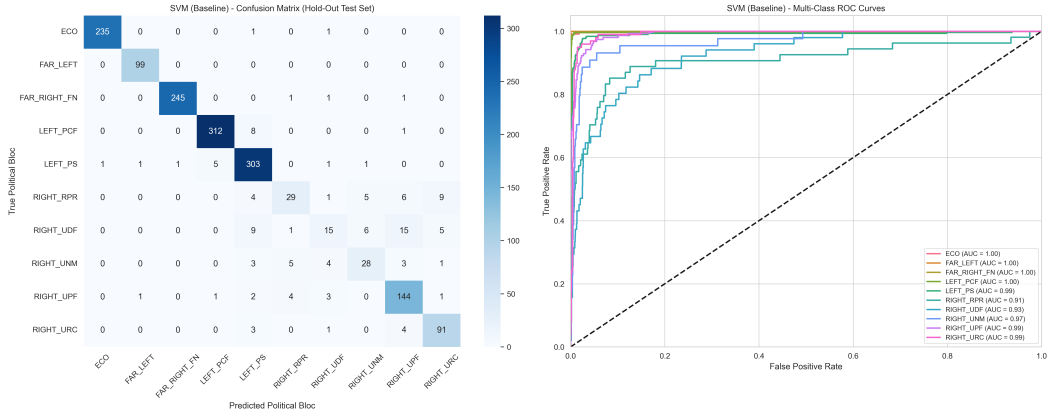


Figure 2: Linear SVM out-of-sample evaluation. Left: Row-normalized empirical confusion matrix M . Right: Multi-class ROC curves demonstrating boundary degradation specifically within the fragmented right-wing and centrist manifolds.

Conversely, the model experiences severe boundary degradation within the mainstream Right. The RIGHT_RPR ($AUC = 0.91$) and RIGHT_UDF ($AUC = 0.93$) exhibit significant off-diagonal density into the coalition classes. For instance, the RPR leaks into the RIGHT_URC ($M_{RPR,URC} = 9$) and RIGHT_UPF ($M_{RPR,UPF} = 6$), while the UDF is equally distributed between its true label and the UPF ($M_{UDF,UPF} = 15$). This is not an algorithmic failure, but a reflection of strict semantic convergence: because the URC and UPF are literal diachronic coalitions of the RPR and UDF, their vocabularies are fundamentally entangled.

Most notably, the confusion matrix exposes a semantic anomaly: 17.6% of the RIGHT_UDF test manifestos are explicitly classified as center-left LEFT_PS ($M_{UDF,PS} = 9$). This suggests that local centrist candidates frequently abandon national right-wing rhetoric in favor of incumbent center-left vocabulary, effectively breaking the sparse linear boundary.

4 Semantic proximity and error topology

4.1 Contextual embedding and non-linear manifold projection (UMAP)

Having established that sparse discriminative boundaries ($\mathbb{R}^{|\mathcal{V}|}$) frequently shatter the true relational proximity of ambiguous manifestos, we shift our objective to mapping the global ideological topology of the corpus. To capture polysemy and syntactic depth, we extract dense, continuous representations using the frozen roberta encoder from our CamemBERT architecture, mapping each document to a vector in $\mathbb{R}^{768}$.

Because analyzing geometric relationships in 768 dimensions is computationally and visually intractable, we apply Uniform Manifold Approximation and Projection (UMAP) McInnes et al. [2018]

to reduce the contextual space to a 2D topological plane ($\mathbb{R}^2$). As detailed in Appendix E, the projection is parameterized using cosine distance to preserve the angular semantic proximity inherent to Transformer embeddings. To quantitatively validate the structural fidelity of this non-linear reduction, we compute the local trustworthiness metric over the out-of-sample projection, achieving a score of $\tau(5) = 0.907$. This confirms a robust preservation of local semantic neighborhoods, though the formal mathematical constraints and geometric limitations of this metric are explicitly detailed in Appendix F.

As illustrated in Figure 3, the resulting 2D manifold refutes the hypothesis of strictly isolated political factions. Instead of discrete clusters, the majority of the corpus converges into a dense central macro-manifold. This centralization reflects the structural reality of the Fifth Republic’s mainstream factions (LEFT_PS, RIGHT_RPR, RIGHT_UDF), which frequently share standardized administrative vocabulary and systemic governance rhetoric.

Conversely, cohesive ideological extremes form peripheral topological clusters. The ECO faction isolates itself at the upper-left quadrant, driven by specific environmental lexicon, while the FAR_RIGHT_FN forms a distinct manifold on the far right of the projection.

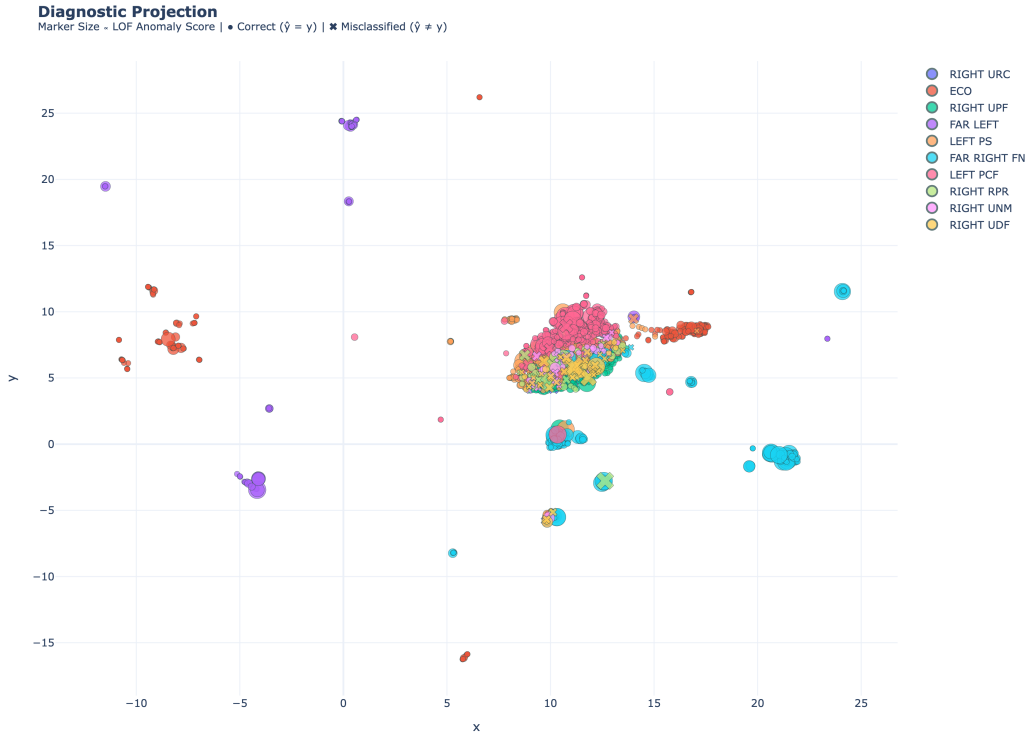


Figure 3: UMAP 2D projection of the $\mathbb{R}^{768}$ CamemBERT test-set embeddings. Misclassifications by the baseline SVM ($\hat{y} \neq y$) are marked with crosses. Marker size is proportional to the Local Outlier Factor (LOF) anomaly score.

Crucially, visual inspection of this projection reveals structural anomalies where manifestos are embedded deep within hostile ideological territories. Several notable visual outliers include:

- EL197_L_1993_03_091_06_1_PF_04 (RIGHT_RPR): Embedded directly within the FAR_RIGHT_FN cluster. This document was misclassified by the baseline models, heavily implying a local Gaullist candidate co-opting extreme-right rhetoric.
- EL195_L_1993_03_075_03_1_PF_04 (LEFT_PS): Projected into the extreme periphery ($x = 21.5, y = -0.89$), far outside the mainstream left manifold, resulting in a model misclassification.

- EL135_L_1981_06_059_20_1_PF_01 (LEFT_PCF): An isolated Communist manifesto ($x = 7.77, y = 9.29$) that was nonetheless correctly classified by the SVM, suggesting its discriminative TF-IDF tokens remained intact despite anomalous contextual phrasing.
- EL174_L_1988_06_015_02_1_PF_04 (LEFT_PCF), EL191_L_1993_03_033_11_2_PF_01 (RIGHT_UPF), and EL135_L_1981_06_050_03_1_PF_02 (LEFT_PS): A triad of competing factions converging into the exact same localized Cartesian space ($x \approx 10.5, y \approx 1.0$), highlighting extreme inter-class semantic overlap.

4.2 Semantic deduplication and statistical anomaly detection ($\mathbb{R}^{768}$)

Visual inspection of a 2D UMAP projection is highly susceptible to geometric distortion, as the algorithm artificially compresses sparse high-dimensional regions to optimize local connectivity. To rigorously cross-validate the heuristic visual anomalies identified in Section 4.1, we perform density-based outlier detection directly on the ambient contextual manifold ($\mathbb{R}^{768}$).

However, prior to density estimation, the manifold must be strictly deduplicated. Due to localized Optical Character Recognition (OCR) artifacts, standard boilerplate manifestos frequently bypass exact string-matching filters. Consequently, these near-duplicates map to coordinates with negligible angular separation ($d_{cos} \approx 0$). In density-based algorithms, this zero-distance collapses the local reachability bounds, causing anomaly scores to artificially explode.

To resolve this, we enforce semantic deduplication via Density-Based Spatial Clustering of Applications with Noise (DBSCAN) Ester et al. [1996]. By parameterizing DBSCAN with a strict radius ($\epsilon = 10^{-4}$) and cosine metric, we isolate and collapse microscopic ϵ -neighborhoods, retaining only a single structural representative per component.

We then compute the Local Outlier Factor (LOF) Breunig et al. [2000] on this sanitized manifold. LOF evaluates the local density deviation of a data point relative to its $k = 15$ nearest neighbors. The model is fit globally over the unified corpus ($\mathcal{D}_{train} \cup \mathcal{D}_{test}$) to establish the baseline structural density of the Fifth Republic’s political discourse. We enforce a strict threshold for statistical anomaly, isolating manifestos where the LOF score exceeds three standard deviations from the global mean ($Z > 3\sigma$). This effectively filters localized semantic variance, extracting only the most extreme structural deviations (e.g., highly idiosyncratic rhetoric or acute topic drift).

The mathematical formalization of both the DBSCAN deduplication prerequisite and the LOF scoring mechanism is detailed in Appendix G.

By cross-referencing the LOF ledger with the previously identified visual anomalies, we confirm that macroscopic spatial isolation in UMAP (e.g., the misclassified LEFT_PS singletons) strongly correlates with elevated LOF scores. However, relying solely on 2D projections remains methodologically insufficient; the highest absolute LOF scores frequently belong to manifestos embedded within seemingly dense peripheral clusters, proving that high-dimensional semantic sparsity is routinely masked by non-linear dimensionality reduction.

4.3 Cross-modal inspection of topological anomalies: OOV control and manifold sparsity

To rigorously interpret the structural anomalies extracted from the deduplicated manifold ($\mathbb{R}^{768}$), we cross-reference the Local Outlier Factor (LOF) scores with the UMAP coordinates ($\mathbb{R}^2$) and the discriminative SVM predictions ($\hat{y}$). Because historical corpora are susceptible to Optical Character Recognition (OCR) corruption, we introduce the Out-Of-Vocabulary (OOV) rate as a strict control variable:

$$\text{OOV} = \frac{|\text{tokens} \notin \mathcal{V}|}{|\text{total document tokens}|} \quad (6)$$

By isolating documents with $\text{OOV} < 0.15$, we guarantee that high anomaly scores are driven by genuine semantic deviation (e.g., idiosyncratic rhetoric, localized topics) rather than character-level noise. Our inspection reveals two distinct geometric phenomena within the corpus:

The False Visual Anomaly ($\mathbb{R}^2$ Distortion): Relying strictly on 2D visual isolation frequently yields false positives. For example, document EL135_L_1981_06_059_20_1_PF_01 (LEFT_PCF) appears highly isolated on the UMAP projection. However, density estimation in the ambient space reveals a completely average structural density ($\text{LOF} = 1.23$, placing it in the 40th percentile of the test set).

This confirms that cross-entropy-based dimensionality reduction can artificially strand normal vectors due to local connectivity optimization, reiterating the necessity of ambient $\mathbb{R}^{768}$ validation.

Extreme Intra-Class Sparsity ($\mathbb{R}^{768}$ Anomalies): Conversely, we identify 35 strict statistical outliers ($Z > 3\sigma$). The distribution of these anomalies is skewed toward specific ideological blocs: FAR_RIGHT_FN ($N = 14$), LEFT_PCF ($N = 6$), and RIGHT_URC ($N = 4$). Strikingly, despite their extreme high-dimensional sparsity, these documents do not trigger boundary misclassifications. For instance, 13 of the 14 FAR_RIGHT_FN anomalies and all 6 of the LEFT_PCF anomalies were correctly classified by the baseline SVM.

This exposes a property of the French political feature space. Blocs like the FAR_RIGHT_FN exhibit intra-class variance, likely driven by highly localized, single-issue campaign rhetoric, resulting in high LOF scores for specific manifestos. However, because their macro-lexicon is fundamentally orthogonal to the mainstream political axes, the global discriminative margin separating them from adjacent classes remains highly robust. They are outliers within their own ideological manifold, yet remain safely bounded away from the hyperplanes of competing factions.

4.4 Generative semantic proximity and isotropic centroid analysis

While the discriminative SVM evaluated boundary separability via sparse token frequencies ($\mathbb{R}^{|\mathcal{V}|}$), it cannot measure continuous ideological proximity. To compute the global semantic similarity between political factions, we analyze the class centroids within the dense CamemBERT representation space ($\mathbb{R}^{768}$).

Because native Transformer embeddings suffer from anisotropy, collapsing the vector space into a narrow conical manifold, cosine similarity on raw hidden states is mathematically degenerate. Therefore, we apply a strict isotropic correction prior to normalization. Let $\mathbf{H} \in \mathbb{R}^{N \times 768}$ be the global embedding matrix. We center the space by subtracting the global mean $\bar{\mathbf{h}}$, and compute the L_2 -normalized empirical centroid $\hat{\mu}_k$ for each ideological class $\mathcal{C}_k$:

$$\hat{\mu}_k = \frac{\sum_{i \in \mathcal{C}_k} (\mathbf{h}_i - \bar{\mathbf{h}})}{\|\sum_{i \in \mathcal{C}_k} (\mathbf{h}_i - \bar{\mathbf{h}})\|_2} \quad (7)$$

We then compute the pairwise semantic proximity matrix via cosine similarity: $\mathbf{S}_{i,j} = \hat{\mu}_i \cdot \hat{\mu}_j$. As illustrated in Figure 4, this continuous projection reveals structural realities obscured by strict linear classification.

The matrix exhibits semantic convergence within the mainstream Right. The RIGHT_UDF and RIGHT_RPR yield a near-identical structural similarity ($\cos(\hat{\mu}_{\text{UDF}}, \hat{\mu}_{\text{RPR}}) = 0.822$). This mathematically corroborates the off-diagonal leakage observed in the baseline confusion matrix (Section 3.6), proving that despite institutional separation, their underlying rhetorical frameworks are functionally indistinguishable. Conversely, ideological hostility is mapped with strict orthogonality, specifically between the FAR_LEFT and RIGHT_URC (-0.580), as well as the LEFT_PS and FAR_RIGHT_FN (-0.540).

Crucially, analyzing continuous centroids resolves a critical paradox regarding the French Left. In the discriminative TF-IDF space, the SVM misclassified LEFT_PCF manifestos almost exclusively into the mainstream LEFT_PS. However, in the continuous generative space, the LEFT_PCF is significantly closer to the FAR_LEFT (0.380) than to the LEFT_PS (0.146).

This exposes the fundamental divergence between sparse and dense NLP architectures: TF-IDF over-fits to pragmatic, localized vocabulary. CamemBERT, however, bypasses local named entities to capture deep syntactic architecture and systemic worldview. While the PCF pragmatically allied with the PS in local elections (tricking the SVM), their continuous semantic topology remains deeply rooted in the Marxist class-struggle dialectics of the Far Left.

Ideological Proximity via Mean-Centered CamemBERT Centroids

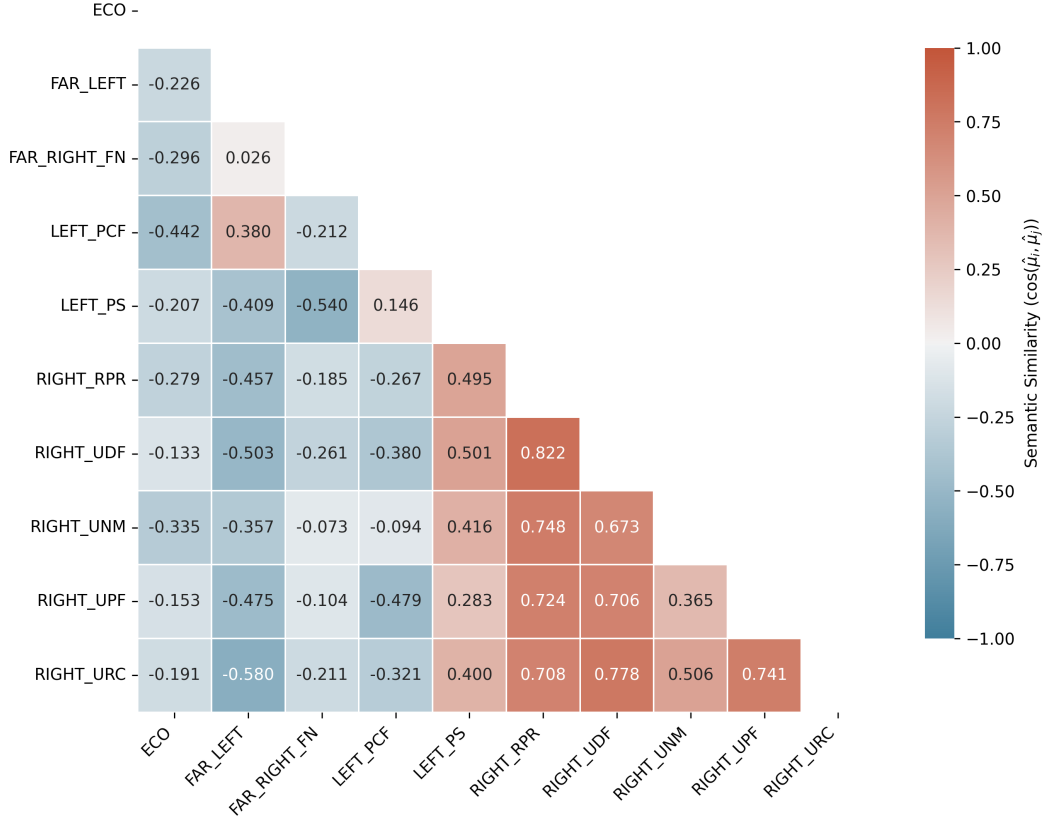


Figure 4: Ideological proximity matrix computed via cosine similarity of zero-mean, L_2 -normalized CamemBERT class centroids.

5 Conclusion

In this study, we established a rigorous NLP pipeline to dissect the structural and ideological topology of the French Fifth Republic’s electoral discourse. By contrasting sparse discriminative optimization with continuous generative manifolds, we exposed the fundamental duality of political text representation.

Our empirical evaluation strictly refutes the default assumption that contextual LLMs uniformly dominate traditional sparse baselines in domain-specific classification. The Linear SVM operating in $\mathbb{R}^{18,054}$ achieved a superior Macro- F_1 of 0.831, outperforming the CamemBERT Linear Probe ($\Delta = +0.109$) and effectively rendering the Zero-Shot NLI architecture obsolete for this localized task. Sparse token frequencies are inherently optimized to exploit the rigid, named-entity-heavy vocabulary required for pragmatic electoral pacts, isolating fragmented macro-blocs with high precision.

However, moving beyond discrete classification, our topological analysis in the $\mathbb{R}^{768}$ CamemBERT space proved that discriminative boundaries obscure latent ideological realities. By applying an isotropic correction to the class centroids, we quantified extreme semantic convergence within the mainstream Right ($\cos(\hat{\mu}_{UDF}, \hat{\mu}_{RPR}) = 0.822$), mathematically justifying their persistent off-diagonal baseline misclassifications. Most critically, the generative space resolved the underlying ideological stance of the LEFT_PCF, demonstrating that despite pragmatic local alliances with the mainstream LEFT_PS, their continuous semantic topology remains strictly anchored to the FAR_LEFT.

Furthermore, our outlier detection methodology established a strict precedent for high-dimensional corpus hygiene. We empirically demonstrated that 2D non-linear projections (UMAP) artificially

manufacture visual anomalies and camouflage true intra-class deviations due to cross-entropy constraints. Robust topological mapping requires explicit deduplication via DBSCAN ($\epsilon = 10^{-4}$) followed by LOF computation strictly within the ambient contextual space.

Limitations and Future Work: We acknowledge two primary architectural constraints in this study. First, the quadratic self-attention mechanism of the roberta encoder strictly bounds the contextual window to 512 tokens, causing structural truncation of extended policy documents. Second, the mapping of highly localized 1980s micro-factions into static macro-blocs introduces subjective label noise. Future work should abandon discrete classification entirely in favor of fully unsupervised, dynamic semantic scaling, utilizing sparse attention mechanisms (e.g., Longformer) to ingest unabridged manifestos and map the continuous drift of French political rhetoric without the constraint of retroactive administrative labels.

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A Pre-processing exclusion criteria and spatial masking

To prevent discriminative models from optimizing on administrative vocabulary or geographical auto-correlation, we hardcoded an exclusion set of domain-specific discourse markers. These tokens represent standard electoral boilerplate ubiquitous across all political factions, providing zero discriminatory variance.

The exact set of excluded lemmas is as follows: *madame, monsieur, mademoiselle, électrice, électeur, concitoyen, cher, chère, salutation, dévoué, fidèle, candidat, circonscription, vote, voter, élection, scrutin, tour, premier, deuxième, suppléant, bulletin, suffrage, député, canton, maire, sortant, mandat, assemblée, vouloir, pouvoir, devoir, faire, falloir, mettre, aller, agir, savoir, répondre, attendre, permettre, croire, ici, oui, non, beaucoup, ensemble, bien, toujours, aujourd'hui, maintenant, ainsi, plus, moins, année, suite, jour, nouveau, nouvelle, fois, moment, cadre, objet, point, particulier, notamment, janvier, février, mars, avril, mai, juin, juillet, août, septembre, octobre, novembre, décembre, lundi, mardi, mercredi, jeudi, vendredi, samedi, dimanche, printemps, été, automne, hiver.*

Furthermore, any token matching an entry in the INSEE COG (Code Officiel Géographique) database for French communes, regions, or departments was dropped, provided it did not overlap with a manually curated list of politically relevant homonyms (e.g., *centre, union, travail*).

A.1 Preprocessing compression example

The following snippet demonstrates the aggressive dimensionality reduction achieved by the pipeline. Administrative boilerplate, geography (“Ain”), dates, and empty rhetoric are stripped, preserving only the dense ideological lemmas.

Raw OCR Text:

ELECTIONS LEGISLATIVES - 14 JUIN 1981 AIN 1e CIRCONSCRIPTION. Micheline ANTONUCCI. POUR QUE ÇA DURE ... La victoire de François Mitterrand, le 10 mai, a exprimé la volonté de changement de la grande majorité des travailleurs, des hommes et des femmes de ce pays. Les élections législatives sont une éta...

Pre-processed Text:

legislative dur exprimer volonte changement grand majorite
travailleur femme pays legislatif important consolidation
majorite gauche adjoint assistant social gauche condition
essentiel reconnaissance pluralisme respecter difference
politique gauche necessaire eviter alliance centriste reduire
entend...

B TF-IDF methodology and parameterization

The sparse baseline representations were constructed using a constrained TF-IDF vectorizer to mitigate the specific linguistic artifacts of French political manifestos.

To dampen the effect of repetitive rhetoric (where a candidate repeats a core campaign word excessively), we applied sublinear term frequency scaling. The raw term frequency $\text{tf}_{i,j}$ (the count of term i in document j) is replaced by its logarithmic mapping:

$$\text{wf}_{i,j} = \begin{cases} 1 + \log(\text{tf}_{i,j}) & \text{if } \text{tf}_{i,j} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

The final weight $w_{i,j}$ applies the Inverse Document Frequency (IDF) and smooths the matrix to prevent division by zero:

$$w_{i,j} = \text{wf}_{i,j} \times \log\left(\frac{N+1}{\text{df}_i+1}\right) + 1 \quad (9)$$

To eliminate OCR noise and administrative boilerplate that escaped the spaCy pipeline, we enforced strict document frequency bounds: terms appearing in fewer than 10 manifestos (min_df) or more than 85% of the corpus (max_df) were discarded. Finally, all document vectors $\mathbf{x}_j$ were L_2 -normalized such that $\|\mathbf{x}_j\|_2 = 1$, ensuring that lengthy presidential-style addresses do not geometrically dominate concise local leaflets within the SVM margin calculations.

To empirically validate the discriminative capacity of the $\mathbb{R}^{18,054}$ feature space prior to SVM optimization, we extract the empirical class centroids. The mean TF-IDF vector μ_k for each ideological macro-bloc k is defined as:

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i \quad (10)$$

By sorting the dimensions of μ_k in descending order, we isolate the highest-weighted uni-grams and bi-grams per faction. As illustrated in Figure 5, this extraction aligns precisely with established French political science heuristics.

The FAR_RIGHT_FN manifold is heavily dominated by anti-immigration and securitarian lexicon (e.g., *immigre*, *immigration*, *insecurite*, *peine*), whereas the FAR_LEFT space is strictly defined by Marxist class-struggle terminology (e.g., *patron*, *bourgeoisie*, *lutte ouvrier*). Furthermore, the presence of the token *socialiste* within the top features of the RIGHT_RPR highlights a critical property of discriminative frequency spaces: high TF-IDF weighting captures not only partisan self-identification but also adversarial rhetorical focus (i.e., the right-wing opposition systematically criticizing the incumbent left-wing government).

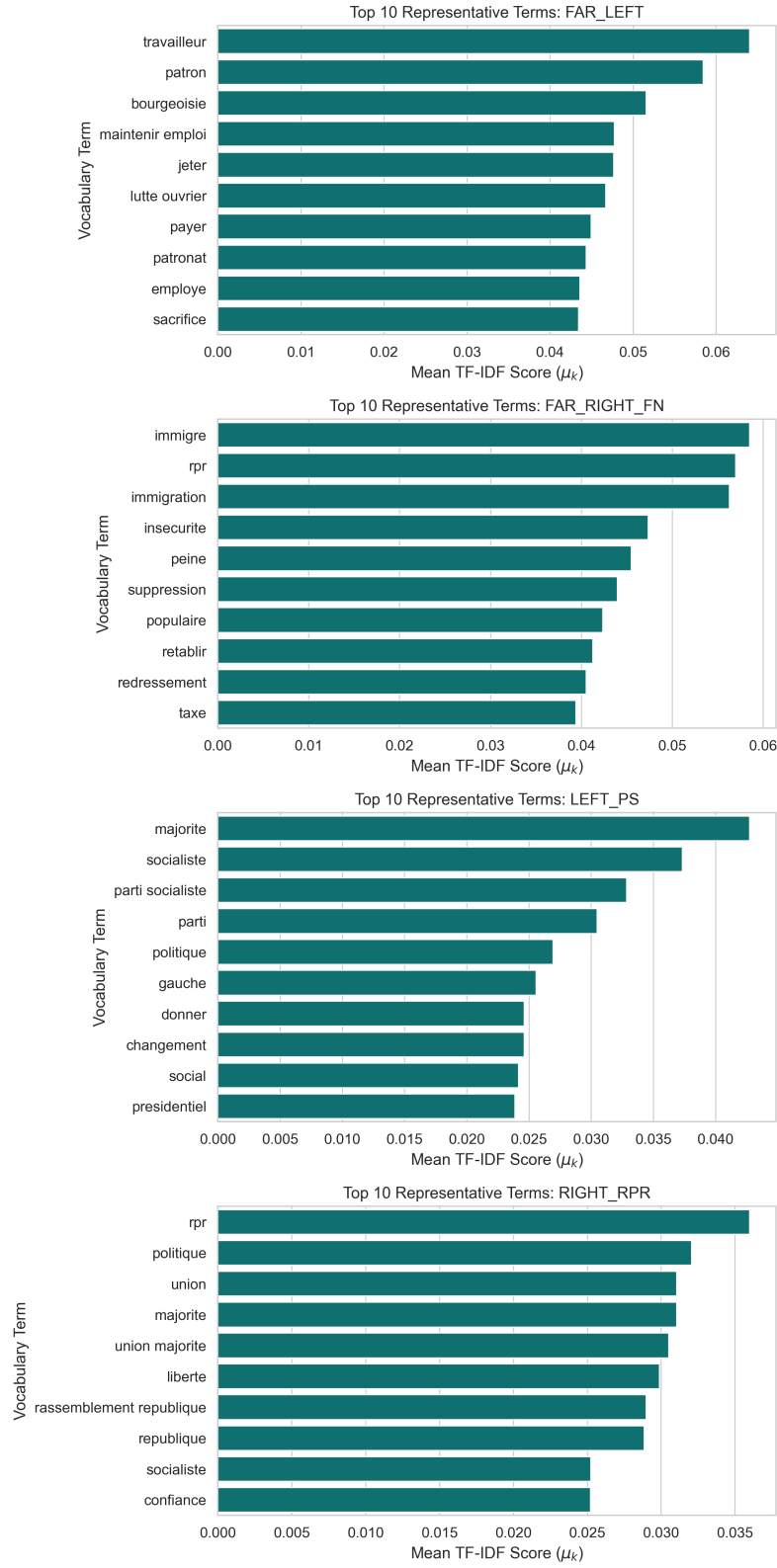


Figure 5: Top 10 representative TF-IDF tokens for selected political factions, derived from the mean dimension weights of the L_2 -normalized document vectors (μ_k).

C Zero-Shot NLI Hypothesis Mapping

In standard multi-class classification, the target space $\mathcal{Y}$ is treated as a set of discrete, semantically opaque indices. However, Zero-Shot inference via Natural Language Inference (NLI) requires these targets to be expressed as semantically rich hypotheses.

To evaluate the mDeBERTa-v3 architecture, we constructed a fixed prompt template: “*Ce manifeste politique soutient [Anchor].*” The precise linguistic anchors used to translate the discrete administrative macro-blocs into continuous contextual representations are detailed in Table 3. These anchors were manually curated to maximize the semantic overlap with standard French political discourse surrounding each faction.

Table 3: Mapping of administrative macro-blocs to NLI linguistic anchors.

Macro-Bloc Label ($\mathcal{Y}$)	NLI Hypothesis Anchor (h_y)
FAR_LEFT	l’extrême gauche et la lutte ouvrière
LEFT_PCF	le Parti Communiste
LEFT_PS	le Parti Socialiste et la gauche
ECO	l’écologie et l’environnement
RIGHT_UNM	la majorité présidentielle de droite
RIGHT_URC	le rassemblement du centre
RIGHT_UDF	l’Union pour la Démocratie Française et le centre droit
RIGHT_RPR	le Rassemblement pour la République et la droite gaulliste
RIGHT_UPF	l’union de la droite
FAR_RIGHT_FN	le Front National et l’extrême droite

D Exhaustive Classification Metrics

Tables 4, 5, and 6 provide the granular, per-class classification reports for the Linear SVM, CamemBERT Linear Probe, and mDeBERTa Zero-Shot models, respectively. Support indicates the absolute frequency of each class in $\mathcal{D}_{test}$.

Table 4: Per-Class Evaluation: TF-IDF + Linear SVM (Baseline)

Ideological Bloc	Precision	Recall	F_1-Score	Support
ECO	0.996	0.992	0.994	237
FAR_LEFT	0.980	1.000	0.990	99
FAR_RIGHT_FN	0.996	0.988	0.992	248
LEFT_PCF	0.981	0.972	0.977	321
LEFT_PS	0.910	0.968	0.938	313
RIGHT_RPR	0.725	0.537	0.617	54
RIGHT_UDF	0.556	0.294	0.385	51
RIGHT_UNM	0.700	0.636	0.667	44
RIGHT_UPF	0.828	0.923	0.873	156
RIGHT_URC	0.850	0.919	0.883	99

Table 5: Per-Class Evaluation: CamemBERT (Linear Probe)

Ideological Bloc	Precision	Recall	F_1-Score	Support
ECO	0.991	0.979	0.985	237
FAR_LEFT	0.980	0.990	0.985	99
FAR_RIGHT_FN	0.996	0.968	0.982	248
LEFT_PCF	0.895	0.953	0.923	321
LEFT_PS	0.840	0.840	0.840	313
RIGHT_RPR	0.308	0.074	0.119	54
RIGHT_UDF	0.483	0.275	0.350	51
RIGHT_UNM	0.473	0.591	0.525	44
RIGHT_UPF	0.812	0.776	0.793	156
RIGHT_URC	0.603	0.889	0.718	99

Table 6: Per-Class Evaluation: mDeBERTa (Zero-Shot NLI)

Ideological Bloc	Precision	Recall	F_1-Score	Support
ECO	0.508	0.979	0.669	237
FAR_LEFT	0.892	0.838	0.865	99
FAR_RIGHT_FN	1.000	0.004	0.008	248
LEFT_PCF	0.732	0.315	0.440	321
LEFT_PS	0.432	0.479	0.455	313
RIGHT_RPR	0.000	0.000	0.000	54
RIGHT_UDF	0.000	0.000	0.000	51
RIGHT_UNM	0.056	0.023	0.032	44
RIGHT_UPF	0.161	0.301	0.210	156
RIGHT_URC	0.304	0.838	0.446	99

E Contextual embeddings and UMAP parameterization

To transition from discrete token counting to continuous semantic modeling, document representations were extracted using the `almanach/camembert-base` transformer. By bypassing the linear classification head and extracting the final hidden state of the [CLS] token, we map each manifesto x_i to a dense vector $\mathbf{h}_i \in \mathbb{R}^{768}$.

To visualize this high-dimensional space, we utilize Uniform Manifold Approximation and Projection (UMAP). UMAP constructs a high-dimensional graph representation of the data using fuzzy simplicial sets and optimizes a low-dimensional graph to be as structurally similar as possible via cross-entropy.

Because magnitude (vector norm) in Transformer embeddings is heavily influenced by sequence length and layer normalization artifacts, Euclidean distance is a sub-optimal metric for semantic similarity. Therefore, we explicitly parameterize UMAP to compute the topological neighborhood using Cosine Distance:

$$d_{cos}(\mathbf{h}_i, \mathbf{h}_j) = 1 - \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{\|\mathbf{h}_i\|_2 \|\mathbf{h}_j\|_2} \quad (11)$$

The algorithm was constrained with $k = 15$ nearest neighbors to balance local and global topological structure, and a minimum projection distance (`min_dist`) of 0.1 to prevent visual over-crowding of densely packed ideological factions. The global manifold was fit on the union of the training and testing sets ($\mathcal{D}_{train} \cup \mathcal{D}_{test}$) to ensure a stable, unified Cartesian space before slicing the $\mathcal{D}_{test}$ coordinates for diagnostic plotting.

F Topological evaluation: Trustworthiness metric

While UMAP excels at mapping low-dimensional structures, non-linear manifold projections inherently distort certain geometric properties of the original feature space. To quantitatively evaluate how well the 2D projection preserves the high-dimensional continuous representations ($\mathbb{R}^{768}$), we compute the trustworthiness metric $\tau(k)$.

Trustworthiness measures the extent to which the k -nearest neighbors of a point in the projected space $\mathbb{R}^2$ correspond to its k -nearest neighbors in the original ambient space $\mathbb{R}^{768}$. It explicitly penalizes points that are placed close together in the low-dimensional projection but were highly distant in the original embeddings (false neighbors). The metric is formalized as:

$$\tau(k) = 1 - \frac{2}{Nk(2N - 3k - 1)} \sum_{i=1}^N \sum_{j \in U_i(k)} (r(i, j) - k) \quad (12)$$

where N is the total number of documents, $U_i(k)$ is the set of points that are among the k -nearest neighbors of document i in the 2D projection but not in the original 768D space, and $r(i, j)$ is the rank of document j among the nearest neighbors of document i in the original space.

For our out-of-sample evaluation, we computed $\tau(k)$ at $k = 5$, yielding a score of 0.907.

Methodological Advantages: Trustworthiness provides a strict, rank-based quantitative measure of local topological preservation. This effectively prevents the reporting of visual artifacts or low-dimensional crowding as genuine semantic clusters, offering a robust defense against the scale distortions typical of cross-entropy-based projections like UMAP.

Geometric Limitations: The metric is mathematically blind to global topological structures. A high $\tau(k)$ score guarantees that local semantic intra-class variance is preserved, but it does not validate the fidelity of macroscopic inter-cluster distances (e.g., the Cartesian distance between the FAR_LEFT and FAR_RIGHT_FN macro-manifolds in the UMAP plot). Furthermore, the metric is highly sensitive to the parameter k ; a projection may perfectly preserve immediate ideological neighbors ($k = 5$) while severely distorting broader structural relationships ($k = 50$).

G Topological deduplication and density-based anomaly scoring

To quantitatively isolate semantic anomalies without the spatial distortion of $\mathbb{R}^2$ projections, we utilize the Local Outlier Factor (LOF). However, the mathematical stability of LOF requires the strict elimination of ϵ -duplicates in the ambient space.

G.1 Pre-processing: DBSCAN ϵ -collapse

Standard textual deduplication is highly vulnerable to OCR noise. Consequently, structurally identical documents yield infinitesimally distinct embeddings $\mathbf{h}_i, \mathbf{h}_j \in \mathbb{R}^{768}$. To prevent these near-duplicates from destabilizing local density calculations, we deploy DBSCAN as a continuous geometric filter prior to LOF.

DBSCAN defines an ϵ -neighborhood around a point $\mathbf{h}_i$ as:

$$N_\epsilon(\mathbf{h}_i) = \{\mathbf{h}_j \in \mathcal{D} \mid d_{\cos}(\mathbf{h}_i, \mathbf{h}_j) \leq \epsilon\} \quad (13)$$

By setting an extremely tight threshold ($\epsilon = 10^{-4}$) and a minimum cluster size of 2, the algorithm identifies microscopic dense components representing structural clones. We retain only the first index of each detected component, projecting the raw dataset into a strictly sanitized, duplicate-free manifold $\mathbf{H}_{\text{unique}}$.

G.2 Local Outlier Factor (LOF) Scoring

Unlike global distance metrics (e.g., Euclidean distance to a cluster centroid), LOF evaluates the *relative* local density of a document’s embedding. The algorithm computes the local reachability density (lrd) of a point A , defined as the inverse of the average reachability distance of its k -nearest neighbors:

$$\text{lrd}_k(A) = \left(\frac{\sum_{B \in N_k(A)} \text{reach-dist}_k(A, B)}{|N_k(A)|} \right)^{-1} \quad (14)$$

Note: Without the prior DBSCAN step, the presence of an identical document B would result in $\text{reach-dist}_k(A, B) = 0$, driving the $\text{lrd}_k(A)$ asymptotically toward infinity and triggering a computational singularity.

The final LOF score compares the local density of A to the local densities of its neighbors:

$$\text{LOF}_k(A) = \frac{\sum_{B \in N_k(A)} \frac{\text{lrd}_k(B)}{\text{lrd}_k(A)}}{|N_k(A)|} \quad (15)$$

An $\text{LOF} \approx 1$ indicates that the document belongs to a region of homogenous density. An $\text{LOF} \gg 1$ mathematically proves that the document is significantly sparser (i.e., semantically isolated) compared to its immediate neighbors.

Methodological Advantages: Because it relies on relative density rather than absolute distance, LOF is uniquely capable of identifying outliers across manifolds of varying variance. A point slightly separated from a highly dense manifold (like the mainstream LEFT_PS) can be correctly flagged as an outlier, whereas a point in a globally sparse cluster (like ECO) will only be flagged if it deviates from that specific local sparsity.

Geometric Limitations and Cross-Validation: The absolute values of LOF are not strictly normalized probabilities, making thresholding heuristic (hence our reliance on a $Z > 3\sigma$ cut-off). Furthermore, analyzing LOF scores explicitly demonstrates the danger of relying strictly on UMAP visual representations.

As illustrated in Figure 6, the absolute top 10 statistical outliers in the ambient space ($\mathbb{R}^{768}$) do not manifest as isolated Cartesian coordinates in the $\mathbb{R}^2$ projection. Instead, they are embedded directly within the densest intra-class macro-manifolds, specifically within the LEFT_PS, RIGHT_UPF, and FAR_RIGHT_FN clusters.

This visual camouflage empirically proves that UMAP’s cross-entropy minimization aggressively compresses points that share a common topological neighborhood, artificially inflating local visual

Test Set Semantic Topology
Highlighting Top 10 Statistical Outliers

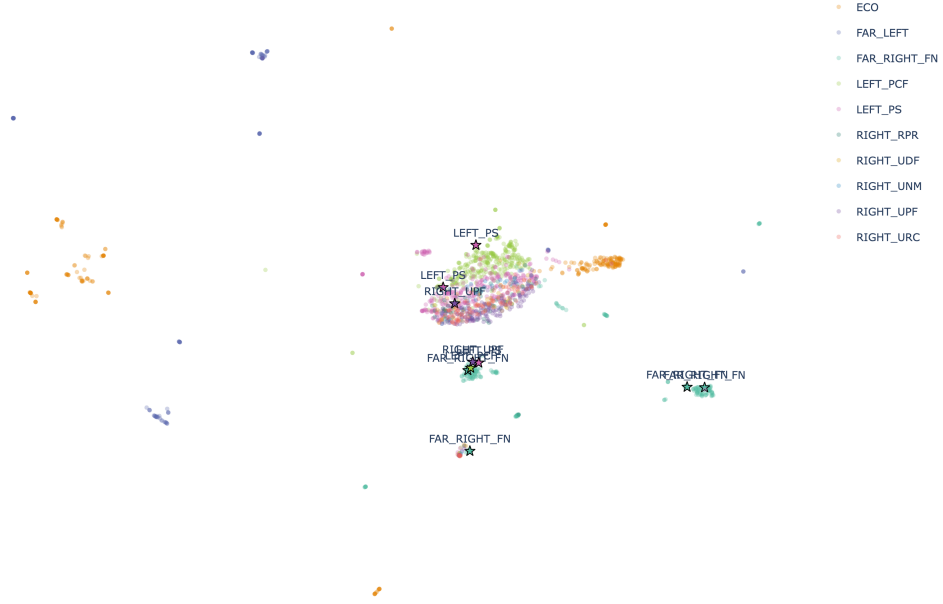


Figure 6: UMAP topology highlighting the absolute top 10 statistical LOF outliers ($Z \gg 3\sigma$) on the deduplicated manifold. Despite their extreme high-dimensional sparsity in $\mathbb{R}^{768}$, the cross-entropy optimization algorithm forces these anomalies into visually contiguous intra-class macro-manifolds.

density. Even if a manifesto exhibits extreme angular deviation from its peers ($d_{cos} \gg \mu_d$), the projection forces it into the aggregate cluster to preserve graph connectivity. Consequently, visual density in lower dimensions is a mathematically deceptive proxy for semantic homogeneity. Robust NLP pipelines must decouple density-based statistical outlier detection (LOF) from non-linear visual dimensionality reduction.